

Curriculum vitae and Track record

Role in the project Project manager ☐ Project partner ☒

Personal information

First name, Surname:	Anders Malthe-Sørenssen		
Date of birth:	13.02.1969	Sex:	Male
Nationality:	Norway		

Education

Year	Faculty/department - University/institution - Country
1998	Ph.D. Department of Physics, University of Oslo, Norway
1994	Master in Physics, Department of Physics, University of Oslo, Norway

Positions - current and previous

Year	Job title – Employer - Country
2023-	Director, Center for Interdisciplinary Education, A Center for Excellence, UiO
2022-2023	Adjunct Chief Research Scientist (20%), Simula research laboratories, Norway
2016-	Director, Center for Computing in Science Education, A Center for Excellence, UiO
2006-	Professor, Department of Physics, University of Oslo
2012-2017	Section Leader, Condensed Matter Physics (10 faculty, 10 post-docs, 30 phd-students), Department of Physics, University of Oslo, Norway
2004-2008	Group leader and educational coordinator, Physics of Geological Processes, a Center of Excellence at University of Oslo
2002-2004	Division lead, Physics of Geological Processes, University of Oslo
2001-2002	Vice President of Technology in Photonyx, a company developing a light modulator for use in fiber optics. Raised 7.5 MUSD of venture capital. Inventor of main technology patent. The company went public in 2003 with a value of 100 MEUR.
1999-2001	Post-doc at University of Oslo, Visiting Scholar at Stanford University
1995-2003	Author of FramView, a simulation tool for physics-based fracture modeling.

Project management experience

Year	Project owner - Project - Role - Funder
Principal investigator (selected recent grants)	
2023-2027	40 Mkr, Center for Interdisciplinary Education, 1 of 3 national centers
2020-2026	8.7 MEUR EU MSCA COFUND, CompSci – Training in Computational Science, with 32 PhD students across 5 disciplines at the University of Oslo.
2019-2022	16.0 Mkr, Bio-inspired neural networks for AI application, NRC IKT+ (10% success rate)
2019-2022	9.2 Mkr, History-dependent friction, NRC-FriPro (5% success rate)
2018-2022	1.5 Mkr, Student research for improved education, Thon Foundation
2017-2021	9.0 Mkr, Structured assessment method, NRC-FinnUt (5% success rate)
2016-2019	1.5 Mkr, Cross-disciplinary research projects for undergraduate students, Thon Foundation

2016-2026	50 Mkr, Center for Computing in Science Education, 1 of 4 national centers.
2015-2019	4.5 Mkr, US-Norwegian collaboration on mineral transf. processes, NRC (25% success rate) a collaboration between Columbia Univ., Univ. of Maryland and USC.
2014-2018	8 Mkr, Coupled Processes in Gas Hydrates, NRC FriPro (5% success rate)
2014-2017	8 Mkr, ShaleSeq - Physiochemistry of CO ₂ sequestration in Pomerian shales, Polish-Norwegian EU project.
2012-2018	8 Mkr, Multiscale physics on the computer, NRC Institute Strategic Project
2011-2015	3.5 Mkr, Numerical Modeling of shale-gas extraction processes, VISTA
2011-2017	10 Mkr, Tight rocks - fundamentals, Statoil, Norway
2010-2016	5.7 Mkr, Permanent CO ₂ storage by in situ injection in ultramafic rocks, NRC Climit
Co-principal investigator (selected recent grants)	
2019-2023	12 Mkr, Artificial Biomimetic systems, UiO:LifeScience convergence project, responsible for 1 postdoc working on machine learning analysis methods.
2019-2023	8 Mkr, CINPLA-2.0, Strategic Faculty Program, University of Oslo
2016-2020	25 Mkr, EarthFlows, Strategic Faculty Project, University of Oslo, responsible for 1 postdoc.
2014-2018	25 Mkr, CINPLA, Center for Integrative Neuroplasticity, Strategic Faculty Program, University of Oslo, responsible for 2 PhD-students.
2016-2020	45 Mkr, DigiBrain, Innovation project from Digital Life Norway directed by Marianne Fyhn, responsible for 1 PhD-student and 1 postdoc.
2016-2021	25 Mkr, BrainMatrix, ToppForsk project directed by Marianne Fyhn, responsible for 1 PhD
2015-2019	8 Mkr, CoBra – computing brain signals, FriPro project directed by Gaute Einevoll.
2013-2017	8 Mkr, CO ₂ /hydrocarb. conversion, FriPro directed by C. Persson, responsible for 1 pd.

Supervision of students (finished/current)

MSc	PhD	Postdocs	University/institution - Country
30/4	18/10	6/4	University of Oslo, Norway

Other relevant professional experiences

Year	Description - Role
2019-2021	Founder and leader, BSc/BA interdisciplinary honours program, Univ. of Oslo
2017-2021	Member, Board of Faculty for Mathematics and Natural Science, Univ. of Oslo
2011-2017	Leader of Bachelor program in Physics (100 students/yr), at the University of Oslo
2013-2019	Member, Program board for Bachelor program in Mathematics, Univ. of Oslo
2012-2013	Fulbright Scholar, Collaboratory for Advanced Computing and Simulations, USC.
2012-2013	Member, Educational Board, Faculty for Education, University of Oslo
2003-2007	Leader, Master program in Physics of Geological Processes, University of Oslo
2016	Elected member of Norwegian Academy of Science and Letters
2013-2016	Editorial Board, Frontiers in Physics
2001-2003	Scientific Advisory Board, Teknoinvest, a major Scandinavian Venture Fund.
2005-2019	Reviewer for Israeli, Dutch and Danish Science Foundations.

Publication statistics

I have published 97 articles in reviewed journals, that have been cited 6248 times (google), with an h-index of 34. I have published 6 articles in Nature (3 in Nature, 2 in Nature Communications, 1 in Nature Geoscience) and 2 articles in PNAS.

I have written an internationally best-selling introductory textbook in physics, “Elementary mechanics using Python” published on Springer that have sold over 300.000 copies since 2015.

10-year track record

My research has focused on fundamental problems in statistical physics, the physics of geological processes (from 1996) – an interdisciplinary program between physics, geoscience, and computer science – with particular focus on forward modeling and machine learning, fundamental studies of friction (from 2009), molecular simulations (from 2012), computational studies of neurophysiological processes (from 2014), and machine learning and AI (from 2022).

Selected publications: Physics of Geological Processes – Mechano-chemical processes

I have built a world-leading activity in geological processes by developing and applying an understanding of geological fracture processes to understand how fracturing is coupled to chemical reactions and fluid transport across scales from 1996 to the present. My research has led to several breakthroughs, e.g. the front page of Nature in 2004.

[1] O. I. Ulven, B. Jamtveit, **A. Malthe-Sørenssen**, Reaction-driven fracturing of porous rock, *Journal of Geophysical Research: Solid Earth* (2014), **119**, 7473-7486 (**49** cit.).

Selected publications: Friction

We have been the first to explain a series of high-impact experimental papers on the onset of sliding friction. Recently, we have introduced a new model for glacier friction, where glacier surges spontaneously emerge in a consistent framework for friction [2], and a new model for history-dependent friction [3].

[2] K. Thøgersen, A. Gilbert, T. V. Schuler, **A. Malthe-Sørenssen**. Rate-and-state friction explains glacier surge propagation. *Nature Communications* (2019). (**66** cit.)

[3] E. Nordhagen, H. A. Sveinsson, **A. Malthe-Sørenssen**. Diffusion-driven frictional aging in silicon carbide, *Tribology Letters* (2023), **71**, 95 (**2** cit.)

Selected publications: Molecular dynamics simulations of geological processes

In 2012 I initiated an activity on the behavior of confined fluids and their importance for geological processes. Example results are that diffusion-osmosis in nano-scale channels may dominate transport during mineral transformation [4] and that the microscopic dynamics of nano-scale facet formation can be studied using molecular dynamics [5]. *In these works, I was responsible for modeling.*

[4] O. Plumper, A. Botan, C. Los, Y. Liu, **A. Malthe-Sørenssen**, B. Jamtveit, Fluid-driven metamorphism of the continental crust governed by nanoscale fluid flow, *Nature Geoscience* (2017) **10**, 685-690 (**118** cit.).

[5] H. A. Sveinsson, A. Hafreager, R. K. Kalia, A. Nakano. P. Vashishta, **A. Malthe-Sørenssen**, Direct atomic simulations of facet formation and equilibrium shapes of SiC nanoparticles, *Crystal Growth and Design* (2020), **20**, 2147-2152. (**10** cit.)

Selected publications: Mechanism of long-term memory formation

In 2013 I co-initiated a collaboration between physicists, biologists and computational scientists to develop an understanding of the molecular mechanisms of long-term memory formation in the brain. An example of an outcome from this collaboration was a spectacular new result in Proceedings of the National Academy of Science [6] and a model for extracellular electrodiffusion [7]. *In these papers I have been responsible for developing the modeling and analysis component.*

[6] E. H. Thompson, K. K. Lensjø, M. B. Wigestrøm, **A. Malthe-Sørenssen**, T. Hafting, M. Fyhn, Removal of perineuronal nets disrupts recall of a remote fear memory, *Proceedings of the National Academy of Sciences* (2018) **115**, 607-612 (**101** cit.)

[7] A. Solbrå, A. Bergersen, J. van den Brink, **A. Malthe-Sørenssen**, G. Einevoll, G. Halnes, A Kirchhoff-Nernst-Planck framework for modeling large scale extracellular electrodiffusion surrounding morphologically detailed neurons, *PLoS comp. biology* (2018), **14**, e1006510 (**28** cit.)

Selected publications: Biologically inspired artificial intelligence

In 2020 I co-initiated a collaboration between physicists, computer scientists and neuroscientists to develop new artificial intelligence methods inspired by biological systems in order to strengthen the coupling

between experimental and computational methods in neuroscience and to make new conceptual advances in our understanding of AI methods.

[8] V. Schøyen, M. B. Pettersen, K. Kolzhausen, M. Fyhn, **A. Malthe-Sørenssen**, M. Lepperød, Coherently remapping toroidal cells but not grid cells are responsible for path integration in virtual agents, *iScience* (2023), 26, 11 (5 cit.)

Selected publications: Machine learning for physical sciences

From 2022 I initiated an activity on machine learning in the physical sciences. Using our unique set of data for frictional surfaces, we demonstrated how to generate surfaces with specific properties using discriminative [9] and generative [10] machine learning models. *I was the leader of these projects.*

[9] F. Najafi, H. A. Sveinsson, C. Dreierstad, H. E. B. Glad, **A. Malthe-Sørenssen**, Modeling the relationship between mechanical yield stress and material geometry using convolutional neural networks, *Applied Physics Letters* (2023), 123, 11 (1 cit.)

[10] E. Nordhagen, H. A. Sveinsson, **A. Malthe-Sørenssen**, Tailoring frictional properties of surfaces using diffusion models, *arXiv* (2024), 2401.05206

Granted patents

The patent [1] formed the basis of Phytonyx, a high-technology company, and describes an electronically controllable thin polymer film that could be used as a tunable, dynamic optical grating. The company went public as Ignis ASA in 2003 with a value of over 200 MEUR and was later bought by a competitor. Several new start-ups have spun off from this company.

[1] **A. Malthe-Sørenssen**, E. Zimmer, T. Natestad, B. Jacobson, Method and device for variable optical attenuator, *US6897995*, May 24, 2005. (25 cit.)

The cross-disciplinary work between physics and geoscience resulted in a patented technology and a software platform for forward modeling of geological systems for use in combination with seismic interpretation and analysis.

[2] **A. Malthe-Sørenssen**, T. Walmann, T. Jossang, J. Feder, H. Hardy, Method of modeling of faulting and fracturing in the Earth, *US7089166*, *US7043410*, *US7031891*, Aug 8, 2006. (78 cit.)

Invited presentations (selected)

From 2018 to 2023 I have held 37 invited talks about our new center of excellence in education, Center for Computing in Science Education, at institutions in Norway, Europe, USA and China, including keynote lectures at international conferences and as part of official Norwegian delegations. In addition, I have been invited to give a wide range of invited talks, such as:

[1] **A. Malthe-Sørenssen**, Friction: behavior across scales, Micro/nano-scale models for tribology, Lorentz Workshop 2017, 30 Jan – 3 Feb, Lorentz Institute, Leiden, Netherlands (2017).

[2] **A. Malthe-Sørenssen**, A minimal model for rupture propagation and its application to front modes, Cecam workshop 2019, 30 Jan – 2 Feb, Lausanne, Switzerland (2019).

Fellowships, awards and prizes

[1] Prize for Best learning environment, University of Oslo, 250kkr (2011 and 2015)

[2] National Prize for excellence in education, Thon Foundation, 500kkr (2015)

[3] Fulbright Scholar, 100kkr (2012)

[4] Prize for Excellence in Education, Ministry of education, Norway, 300kkr (2012)

[5] Nansen Prize for Young Researchers, Norw. Acad. of Science and Letters, 100kkr (2001)

[6] His Majesty the King's Gold Medal for best PhD-thesis in science, UiO (1999)

Major contributions to the early careers of excellent researchers

[1] Adriano Mazzini, my post-doc on mud volcanoes from 2004 to 2007 and from 2010 to 2011, received an **ERC Starting grant** on “Lusi: a unique natural laboratory for multidisciplinary studies of focused fluid flow in sedimentary basins”, 2013-2017. We have several joint publications from this period, including A. Mazzini, H. Svensen, G. G. Akhmanov, G. Aliosi, S. Planke, **A. Malthe-Sørenssen**. Triggering and dynamic evolution of the LUSI mud volcano, Indonesia, *Earth and Planetary Science Letter* (2007) **261**, 375–388, (300 cit.).

[2] Phd-student Jørgen Trømborg received HMK Gold Medal for best PhD thesis in Science at UiO in 2016.